

Section 1 -Getting Started

2026年6月1日 9:46

Why would you use TypeScript?

The screenshot shows the TypeScript website homepage. A pink box highlights the text "TypeScript is JavaScript with syntax for types." with a pink arrow pointing to it. Below this, it states "TypeScript is a strongly typed programming language that builds on JavaScript, giving you better tooling at any scale." and includes a "Try TypeScript Now" button. On the right, there is a code editor snippet showing a TypeScript object and a console log, with a red error message: "Property 'name' does not exist on type '{ firstName: string; lastName: string; role: string; }'." A dark overlay at the top right says "Official TypeScript website: typescriptlang.org." The bottom of the page mentions "TypeScript 5.5 is now available, 5.6 is currently in beta."

TypeScript

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TypeScript is JavaScript with syntax for types.

TypeScript is a strongly typed programming language that builds on JavaScript, giving you better tooling at any scale.

Try TypeScript Now
Online or via npm

Official TypeScript website: typescriptlang.org.

```
const user = {
  firstName: "Angela",
  lastName: "Davis",
  role: "Professor",
}

console.log(user.name)
```

Property 'name' does not exist on type '{ firstName: string; lastName: string; role: string; }'.

TypeScript 5.5 is now available, 5.6 is currently in beta.

What is TypeScript?

JavaScript and More

A Result You Can Trust

Safety at Scale

TS calculator.ts ✕

```
1 function deriveFinalPrice(inputPrice: number) {
2   const finalPrice = inputPrice + inputPrice * 0.19;
3   const outputEl = document.getElementById('final-price')!;
4   outputEl.textContent = 'Final Price: ' + finalPrice + ' €';
5 }
6
7 const formEl = document.querySelector('form')!;
8
9 formEl.addEventListener('submit', function (event) {
10   event.preventDefault();
11   const fd = new FormData(event.currentTarget as HTMLFormElement);
12   const inputPrice = fd.get('price')!;
13   deriveFinalPrice(+inputPrice);
14 });
```

TypeScript is a JavaScript superset

It's an "extension" to the JavaScript language

The core syntax & features are the same
but extra features are added

Most importantly: Strict & static typing

2. Why would you use TypeScript?

Why would you use TypeScript?

I have prepared a project

```
Understanding TypeScript

JS calculator.js M X

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15
```

TypeScript - What & Why?

PRICE

Calculate Final Price

Final Price: 152.85 €

JS calculator.js M X

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15
```

I'm having some problems with the types

Problems could have been avoided if we were using typescript.

app.css
calculator.ts
index.html



with a calculator.ts file instead of js.

calculator.ts ×

```
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4      outputEl.textContent = 'Final Price: ' + finalPrice + ' €';  
5  }  
6  
7  const formEl = document.querySelector('form')!;  
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9  formEl.addEventListener('submit', function (event) {  
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13     deriveFinalPrice(+inputPrice);  
14 });  
15
```

❏

TS calculator.ts 1 ●

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15
```

> 2.85 + "15"

< '2.8515'



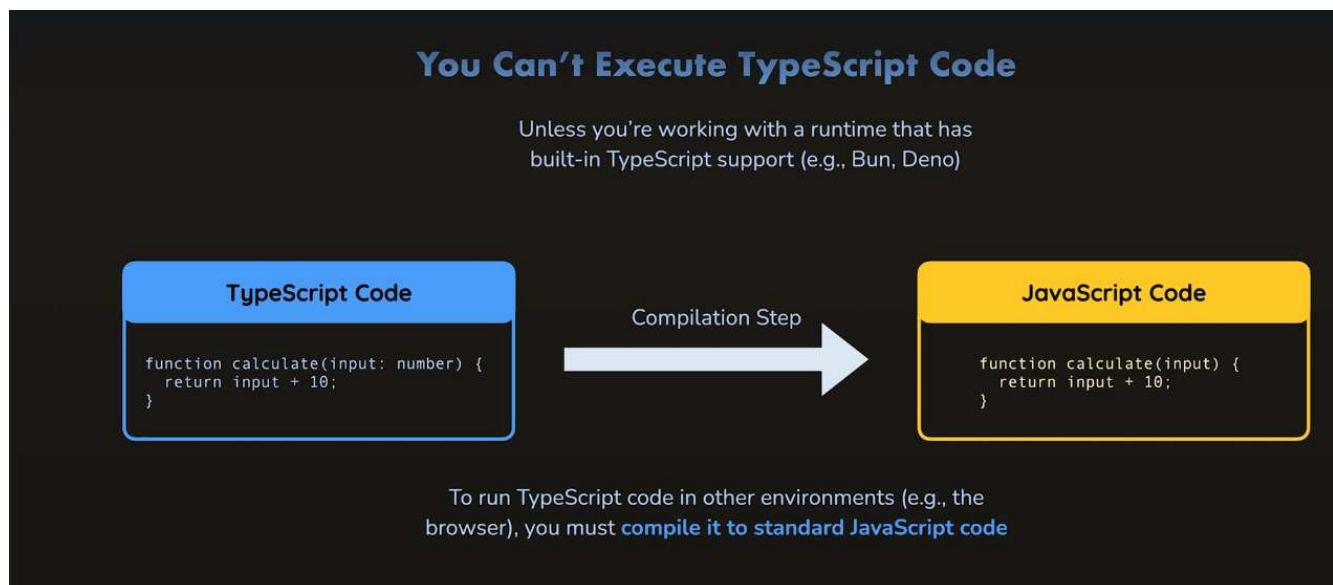
>

and produce a longer string where it simply concatenates

Installing and Using TypeScript

This typescript code does not run on the browser.

```
TS calculator.ts X
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```



That's why in the official doc of typescript, you need to download a compiler that will compile your typescript code to javascript code.

Download TypeScript

TypeScript can be installed through three installation routes depending on how you intend to use it: an npm module, a NuGet package or a Visual Studio Extension.

If you are using Node.js, you want the npm version. If you are using MSBuild in your project, you want the NuGet package or Visual Studio extension.

TypeScript in Your Project

Having TypeScript set up on a per-project basis lets you have many projects with many different versions of TypeScript, this keeps each project working consistently.

via npm

TypeScript is available as a [package on the npm registry](#), available as `"typescript"`.

You will need a copy of [Node.js](#) as an environment to run the package. Then you use a dependency manager like [npm](#), [yarn](#) or [pnpm](#) to download TypeScript into your project.

```
npm install typescript --save-dev
```

`npm` `yarn` `pnpm`

All of these dependency managers are **a tool that will convert the TypeScript code** to JavaScript, and then running.

with Visual Studio

For most project types, you can get TypeScript as a package in NuGet for your MSBuild projects, for example an ASP.NET Core app.

When using NuGet, you can [install TypeScript through Visual Studio](#) using:

- The Manage NuGet Packages window (which you can get to by right-clicking on a project node)
- The NuGet Package Manager Console (found in Tools > Package Manager Console) and then running:

Globally Installing TypeScript

It can be handy to have TypeScript available across all projects, often to test one-off ideas. Long-term, codebases should prefer a project-wide installation over a global install so that they can benefit from reproducible builds across different machines.

via npm

You can use npm to install TypeScript globally, this means that you can use the `tsc` command anywhere in your terminal.

To do this, run `npm install -g typescript`. This will install the latest version (currently 5.5).

via Visual Studio Marketplace

You can install TypeScript as a Visual Studio extension, which will allow you to use TypeScript across many MSBuild projects in Visual Studio.

The latest version is available [in the Visual Studio Marketplace](#).

We will install via npm.

Run JavaScript Everywhere

Node.js® is a free, open-source, cross-platform JavaScript runtime environment that lets developers create servers, web apps, command line tools and scripts.

[Download Node.js \(LTS\)](#)

Downloads Node.js v20.16.0 with long-term support. Node.js can also be installed via package managers.

Want new features sooner? Get Node.js v22.6.0 instead.

Create an HTTP Server Write Tests Read and Hash a File Streams Pipeline Work with

```
1 // server.mjs
2 import { createServer } from 'node:http';
3
4 const server = createServer((req, res) => {
5   res.writeHead(200, { 'Content-Type': 'text/plain' });
6   res.end('Hello World!\n');
7 });
8
9 // starts a simple http server locally on port 3000
10 server.listen(3000, '127.0.0.1', () => {
11   console.log('Listening on 127.0.0.1:3000');
12 });
13
14 // run with 'node server.mjs'
```

JavaScript

[Copy to clipboard](#)

[Learn more what Node.js is able to offer with our Learning materials.](#)

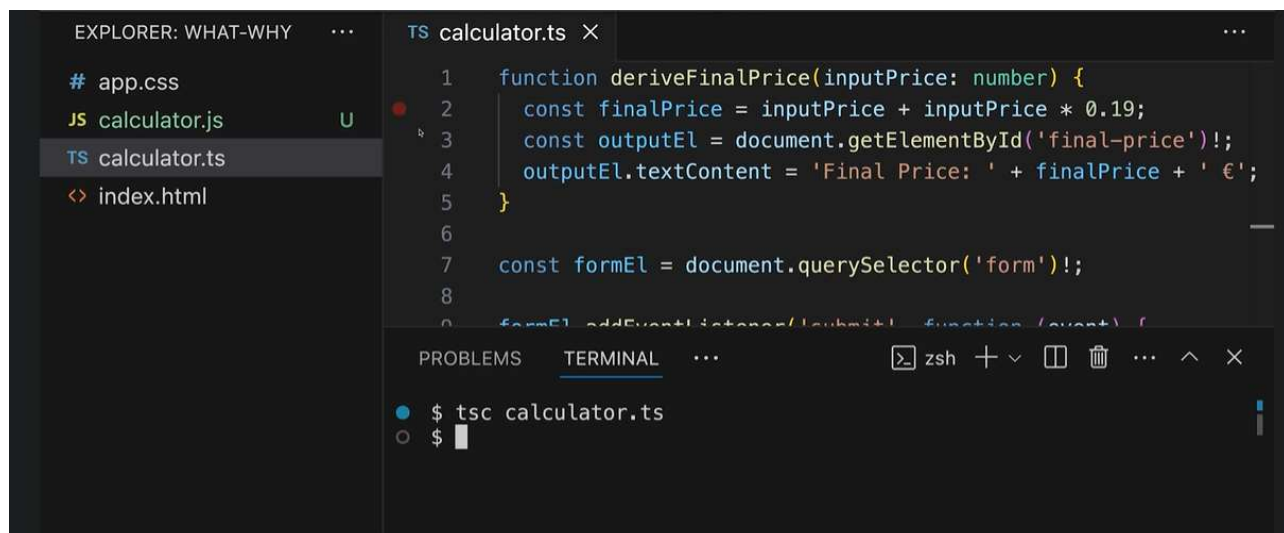
And npm is a package manager

Npm is a package manager that comes together with Node.JS.

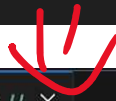
Now open up the terminal and type

'tsc calculator.ts'

And you will see the .js file



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 **Visual Studio Code**
Software

Visual Studio Code, commonly referred to as VS Code, is an integrated development environment developed by Microsoft for Windows, Linux, macOS and web browsers. Features include support for debugging, syntax highlighting, intelligent code ... [More](#)

[Source: Wikipedia >](#)

Initial release date: April 29, 2015

Programming languages: JavaScript, C, C#, CSS, TypeScript

Developer: Microsoft

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